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ORIGINAL ARTICLE

Intensive Versus Conventional Glycemic Control in Patients with Diabetes During Enteral Nutrition After Gastrectomy

Jiaxiang Yuan[1] · Tao Liu[2] · Xiefu Zhang[2] · Yaqing Si[1] · Yanwei Ye[2] · Chunlin Zhao[2] · Qun Wang[1] · Xinsheng Shen[1]

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Abstract

Background This study compared intensive and conventional glycemic management strategies in diabetic patients receiving enteral nutrition after gastrectomy.

Methods Diabetic patients (n=212) who underwent gastrectomy between September 2006 and March 2014 were randomized to intensive glycemic (IG) management with continuous insulin infusion (target glucose 4.4–6.1 mmol/l (80–110 mg/dl)) or conventional glycemic (CG) management with intermittent bolus insulin (target glucose <11.1 mmol/l (<200 mg/dl)). Outcomes included blood glucose concentrations, insulin administration, and postoperative morbidity and mortality.

Results Blood glucose levels were lower (5.4±1.2 vs. 9.5±1.8 mmol/l, P<0.001) and mean insulin dose was higher (55±15 vs.32±16 units/day, P<0.001) in the IG than in the CG group. Rates of severe hypoglycemia (7.5 vs. 0.9 %, P=0.035) and achievement of target blood glucose (86.3 vs. 72.6 %, P=0.023) were higher, while severe hyperglycemia rate was lower (1.9 vs. 11.3 %, P=0.010), in the IG group. Surgical site infection rate was lower in the IG group (4.7 vs. 13.2 %, P<0.030). Rates of other infective complications, bleeding, delayed gastric emptying, obstruction, hepatic dysfunction, renal dysfunction, and circulatory insufficiency were similar in the two groups.

Conclusions Intensive glycemic control in diabetic patients receiving enteral nutrition after gastrectomy was associated with a lower surgical site infection rate but a higher hypoglycemia rate.

Keywords Glycemic control · Enteral nutrition · Diabetes · Gastrectomy

Introduction

Various benign and malignant tumors can develop in the stomach. Although there have been many advances in the treatment of gastric tumors, gastrectomy remains the mainstay of radical curative treatment. Unfortunately, gastrectomy is frequently associated with postoperative morbidity and mortality, especially in patients with diabetes.[1, 2]

Hyperglycemia is part of the body's stress response to surgery, and gastrectomy has been found effective at controlling glycemia in overweight patients with type 2 diabetes.[3, 4] Early enteral nutrition after upper gastrointestinal surgical resection has been associated with a significantly shortened length of hospital stay and improved clinical outcomes.[5–7] Because of the inability to assess the exact amount of glucose absorbed through the intestines, it is difficult to control blood glucose in patients receiving enteral nutrition. This study assessed whether intensive glycemic control was well-tolerated, safe, and improved clinical outcomes in diabetic patients receiving enteral nutrition after gastrectomy.

Jiaxiang Yuan and Tao Liu contributed equally to this work.

✉ Xiefu Zhang
zhangxiefudoctor@163.com

[1] Department of Laparoscopic Surgery, The First Affiliated Hospital of Zhengzhou University, Zhengzhou, China

[2] Department of Gastrointestinal Surgery, The First Affiliated Hospital of Zhengzhou University, No. 1 East Jianshe Road, Zhengzhou, Henan, China

Patients and Methods

Patients and Design

This study was approved by the Ethical Committee of the First Affiliated Hospital of Zhengzhou University, and written informed consent was obtained from each patient or from a legal surrogate. The study cohort consisted of adult patients with type 2 diabetes mellitus (DM) undergoing gastrectomy for gastric tumors between September 2006 and March 2014. Type 2 DM was diagnosed according to the criteria of the American Diabetes Association and gastric tumors were diagnosed endoscopically or by imaging modalities before surgery. Patients with gastric cancer underwent radical D2 resection, preserving the pancreas and spleen whenever possible. Patients with other gastric tumors, including gastrointestinal stromal tumors, malignant lymphomas, and leiomyomas, underwent partial or total gastrectomy. The type of surgery was determined by the anatomical location of the tumor. At the end of surgery, a naso-jejunal tube was inserted into the jejunum 30 cm from the gastrojejunal/esophagojejunal anastomosis or resection site.

Patients were randomly assigned to receive intensive glycemic (IG) management, with continuous insulin infusion to a target blood glucose concentration 4.4–6.1 mmol/l (80–110 mg/dl), or conventional glycemic (CG) management, with intermittent bolus insulin to a target blood glucose concentration <11.1 mmol/l (<200 mg/dl). Patients in the IG group were started on intravenous infusion of 0.5–1 U/h insulin. Blood glucose was monitored hourly (every 2–4 h when stable), with the insulin infusion rate adjusted according to an algorithm (Table 1). Patients in the CG group were administered insulin subcutaneously every 4–6 h, based on the results of bedside glucose monitoring, with extra injections administered if necessary (Table 2).

Table 1 Intensive group intravenous insulin algorithm

Current blood glucose (mmol/l)	Action
>12.0	Increase insulin infusion rate by 4 units/h Give intravenous bolus of insulin 2 units Call physician
10.0–12.0	Increase insulin infusion rate by 3 units/h
8.0–10.0	Increase insulin infusion rate by 2 units/h
6.1–8.0	Increase insulin infusion rate by 1 units/h
4.4–6.1	Maintain current insulin infusion rate Ensure basic glucose intake
2.2–4.4	Stop insulin infusion Ensure basic glucose intake Give bolus of 10 ml 50 % glucose Call physician and recheck
<2.2	Stop insulin infusion Ensure basic glucose intake Give bolus of 20 ml 50 % glucose Call physician and recheck

Mix 100 U short-acting insulin in 100 ml normal saline

Postoperative Management and Data Collection

Patients were infused with 250 ml of normal saline, starting within 12 h after surgery. Patients received feedings of 20 ml/h SP or TPF (Nutricia) through a naso-jejunal tube beginning on the first postoperative day, with the rate increasing 10 ml/h as tolerated every 12–24 h, to a maximum rate of 80 ml/h.[8] The average caloric intake was 25–30 kcal/kg/day. Between the eighth and tenth days, the naso-jejunal tube was removed except for special reasons. Patients were excluded if (1) a withdrawal request was made by the patient or surrogate; (2) the patient underwent laparotomy or palliative surgery; (3) the patient was unable to tolerate enteral nutrition, as shown by vomiting, diarrhea, or abdominal distention; or (4) the naso-jejunal tube became occluded or was pulled out.

Blood glucose concentrations, insulin administration, volume of enteral and parenteral nutrition, and additional intravenous glucose administered were recorded. Serious adverse events included severe hypoglycemia, defined as a blood glucose level ≤2.2 mmol/l (≤40 mg/dl), and severe hyperglycemia, defined as a blood glucose level ≥16.7 mmol/l (≥300 mg/dl). Postoperative surveillance included clinical examinations and laboratory tests. Outcome measures included postoperative morbidity and mortality rates.

Statistical Analysis

Continuous variables are presented as mean±SD, with normally distributed data compared using Student's *t*

Table 2 Conventional group insulin algorithm

Current blood glucose (mmol/l)	Action
>22.0	Administered insulin subcutaneously 12 units Call physician
20.0–22.0	Administered insulin subcutaneously 10 units
17.0–20.0	Administered insulin subcutaneously 8 units
14.0–17.0	Administered insulin subcutaneously 6 units
11.1–14.0	Administered insulin subcutaneously 4 units
4.4–11.1	Ensure basic glucose intake
2.2–4.4	Ensure basic glucose intake Give bolus of 10 ml 50 % glucose Call physician and recheck
<2.2	Ensure basic glucose intake Give bolus of 20 ml 50 % glucose Call physician and recheck

Table 3 Baseline characteristics and sugical data of patients

Variables	Intensive group	Conventional group
Gender (f/m)	60/46	65/41
Age (years)	60.5±13.2	61.1±13.5
Type of tumor		
Gastric cancer	72	70
Ectopic pancreatic tissue	2	4
Carcinoids (neuroendocrine tumors)	3	5
Gastrointestinal stromal cell tumors	25	24
Lipomas	4	3
Albumin (g/l)	33.2±3.8	32.8±3.6
Duration of surgery (min)	140.6±28.7	132.8±30.5
Operative blood loss (ml)	230±80	250±73
HbA1c (%)	8.0±0.6	7.9±0.5
BMI (kg/m^2)	22.1±2.5	21.6±2.0
Blood glucose level (mmol/l)	8.5±2.4	8.2±2.6
DM duration (years)	6.2±0.4	5.8±0.7
Blood transfusion, *n* (%)	22 (20.8)	20 (18.9)
Comorbidity, *n* (%)		
Respiratory disfunction	6 (5.7)	8 (7.5)
Cardiovascular disfunction	16 (15.1)	15 (14.2)
Liver disfunction	3 (2.8)	4 (3.8)
Renal disfunction	1 (0.9)	2 (1.9)
Cerebrovascular disease	2 (1.9)	2 (1.9)
Type of operation, *n* (%)		
Local gastrectomy	21 (19.8)	26 (24.5)
Distal gastrectomy	40 (37.7)	38 (35.8)
Proximal gastrectomy	22 (20.8)	25 (23.6)
Total gastrectomy	23 (21.7)	17 (16.0)
Location of tumor, *n* (%)		
Upper third	22 (20.8)	26 (24.5)
Middle third	18 (17.0)	19 (17.9)
Distal third	54 (50.9)	52 (49.1)
More than two thirds	12 (11.3)	9 (8.5)
Treatment for DM, *n* (%)		
Insulin	71 (67.0)	67 (63.2)
Oral agents	25 (23.6)	30 (28.3)
None	10 (9.4)	9 (8.5)
ASA (1/2/3), *n* (%)	23/68/15 (21.7/64.2/14.2)	25/62/19 (23.6/58.5/17.9)

Data are given as mean±SD, median, or number (%)

BMI body mass index, *DM* diabetes mellitus, *Hb1Ac* glycolsylated hemoglobin, *FBG* fasting blood glucose, *ASA* American Society of Anesthesiologists classification

tests, non-normally distributed data compared using Mann–Whitney *U* tests, and proportions compared using χ^2 or Fisher's exact test, as warranted. A *P* value <0.05 was considered statistically significant. All analyses were performed using SPSS software (SPSS, Chicago, IL, USA).

Results

Study Participants and Surgical Data

Of the 248 patients with type 2 DM who underwent gastrectomy between September 2006 and March 2014 at the First

Table 4 Postoperative outcomes and adverse events

Outcomes and adverse events	Intensive group	Conventional group	*P* value
Blood glucose level (mmol/l)	5.4±1.2	9.5±1.8	<0.001
Severe hypoglycemia	8 (7.5 %)	1 (0.9 %)	0.035
Severe hyperglycemia	2 (1.9 %)	12 (11.3 %)	0.010
Achieved target blood glucose level	86.3 %	72.6 %	0.023
Insulin dose (units/day)	55±15	32±16	<0.001
Mortality	1 (0.9 %)	1 (0.9 %)	1
Infective complications			
Surgical site	5 (4.7 %)	14 (13.2 %)	0.030
Pneumonia	6 (5.7 %)	8 (7.5 %)	0.580
Urinary tract	7 (6.6 %)	6 (5.7 %)	0.775
Bacteremia	3 (2.8 %)	4 (3.8 %)	1
Anastomotic leak	2 (1.9 %)	3 (2.8 %)	0.685
Non-infective complications			
Bleeding	0	1 (0.9 %)	1
Delayed gastric emptying	5 (4.7 %)	7 (6.6 %)	0.552
Obstruction	1 (0.9 %)	2 (1.9 %)	1
Hepatic dysfunction	6 (5.7 %)	6 (5.7 %)	1
Renal dysfunction	0	1 (0.9 %)	1
Circulatory insufficiency	3 (2.8 %)	1 (0.9 %)	0.621

Affiliated Hospital of Zhengzhou University, 36 were excluded, with the remaining 212 patients randomized to the IG or CG group. Of these 212 patients, 142 had been diagnosed with gastric cancer and 70 with other tumors. There were no significant differences in the characteristics and surgical data of the IG and CG groups (Table 3). Stages of gastric cancers and types of tumors were similar in the two groups.

Outcomes

Table 4 shows the postoperative outcomes and adverse events for the two groups. Blood glucose levels were significantly lower (5.4±1.2 vs. 9.5±1.8 mmol/l, $P<0.001$) (Fig. 1), and mean insulin dose was significantly higher (55±15 vs.32±16 units/day, $P<0.001$), in the IG than in the CG group. Severe hypoglycemia was recorded in eight patients (7.5 %) in the IG group compared with one (0.9 %) in the CG group ($P=0.035$). Because the study design was unblinded, allowing appropriate treatment, there were no long-term sequelae of severe hypoglycemia. The percentage of patients experiencing severe hyperglycemia was significantly lower in the IG than in the CG group (1.9 vs. 11.3 %, $P=0.010$), and patients in the IG group were significantly more likely to achieve target blood glucose levels (86.3 vs. 72.6 %, $P=0.023$).

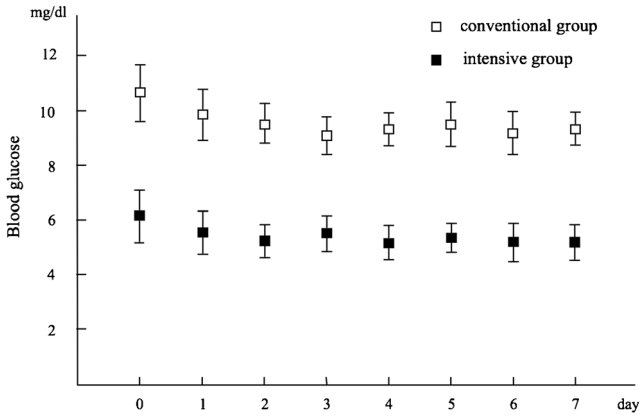

Fig. 1 Time independence of changes in postoperative blood glucose concentrations in IG and CG. Data are mean blood glucose concentration±SD (mg/dl)

During the study, one patient in each group died; the causes of death were acute myocardial infarction in the IG group patient and cerebrovascular accident in the CG group patient. The rate of infection at the surgical site was significantly lower in the IG than in the CG group (4.7 vs. 13.2 %, $P=0.030$). The

Table 5 Percentages of patients with surgical site infection from 2006 to 2014

Year	Intensive group	Conventional group
2006	1/7 (14.3 %)	0/8
2007	0/8	2/9 (22.2 %)
2008	1/10 (10 %)	1/8 (12.5 %)
2009	0/12	0/10
2010	0/15	3/16 (18.8 %)
2011	1/13 (7.7 %)	1/12 (8.3 %)
2012	0/14	2/13 (15.4 %)
2013	1/14 (7.1 %)	2/13 (15.4 %)
2014	1/13 (7.7 %)	3/17 (17.6 %)

percentages of patients with surgical site infection for both IG and CG groups each year from 2006 to 2014 were listed in Table 5. The rates of other infective complications, bleeding, delayed gastric emptying, obstruction, hepatic dysfunction, renal dysfunction, and circulatory insufficiency were similar in the two groups.

Discussion

This study showed that intensive glycemic control, when compared with conventional glycemic control, in patients with type 2 DM receiving enteral nutrition after gastrectomy, was associated with significantly reduced surgical site infection and increased risk of hypoglycemia. The two management systems, however, did not differ in rates of other infective and non-infective complications and mortality. Patients in the IG group were more likely to achieve target blood glucose concentrations, with lower rates of severe hyperglycemia, than patients in the CG group.

Hyperglycemia is a common response to perioperative stress. Surgery and anesthesia invoke stereotypical alterations in carbohydrate metabolism, including increased glucose production and impaired insulin secretion and sensitivity. Hyperglycemia may cause many adverse effects, such as considerable cellular dehydration, osmotic diuresis, increased inflammation, vulnerability to infection, and damage to various tissues.[9] Although intensive glycemic control has been reported to reduce morbidity and mortality rates in critical ill and surgical patients,[1, 10] other trials did not confirm these results, indicating that intensive glycemic control increased the rates of hypoglycemia and mortality.[11–13] Our study found that the surgical site infection rate was significantly lower (4.7 vs. 13.2 %, P=0.030) and the severe hypoglycemia rate was significantly higher (7.5 vs. 0.9 %, P=0.035) in the IG than in the CG group. Unlike the findings in critically ill patients, this study found no difference in mortality rates.[12] However, the low mortality rates in both groups indicate that studies in larger patient samples are needed for adequate statistical comparisons.

A unique feature of our study was starting enteral nutrition soon after gastrectomy. Early enteral nutrition plays an important role in patients undergoing major upper gastrointestinal resection.[8] Insulin secretion is controlled almost entirely by blood glucose concentration. However, gastrointestinal hormones, including gastric inhibitory peptide, gastrin, secretin, and cholecystokinin, induce a moderate increase in insulin secretion as blood glucose concentrations increase. These hormones also increase the sensitivity of the insulin response to increased blood glucose.[14, 15] Early enteral nutrition has been reported to alleviate insulin resistance following surgery in gastric cancer patients.[16] Moreover, the rate of severe hypoglycemia in these patients was lower than in other trials,[17] suggesting that incretin hormones play a significant role in regulating glucose homeostasis.

Patients with gastric tumors not only present with significant malnutrition but also receive parenteral nutrition after gastrectomy, making it difficult to control their blood glucose concentrations. This trial showed that patients in the IG group, receiving continuous insulin infusions, were more likely to achieve target blood glucose levels than patients in the CG group. Unfortunately, IG was also associated with a significantly increased occurrence of hypoglycemia. In patients undergoing isolated coronary artery bypass grafting, moderate glycemic control, to a target of 7.1–9.9 mg/dl, was found to be superior to tight glycemic control, to a target of <7.0 mg/dl, with decreased rates of mortality, major complications, and hypoglycemic events.[18] More studies evaluating optimal postoperative glycemic regimens are necessary.

The limitations of this trial include it not having a strictly unblinded design and its performance at a single center. Because patients who experience severe hypoglycemia should receive rapid and appropriate treatment, clinicians should be informed that these patients, especially those receiving intensive glycemic control, are prone to hypoglycemia. Therefore, these results require verification by larger, double-blind, multi-center studies. In this trail, all staff members monitored blood glucose and administered insulin according to strict protocols. To minimize bias, the 212 patients were randomized over 8 years, with none of the clinicians aware of the purpose of this study.

Conclusion

This study showed that intensive glycemic control, targeting a blood glucose level of 4.4–6.1 mmol/l, resulted in a lower surgical site infection rate than conventional glycemic control, targeting a blood glucose level of <11.1 mmol/l, in patients with type 2 DM receiving enteral nutrition after gastrectomy. Intensive glycemic control, however, was associated with a significantly increased risk of hypoglycemia. New technologies, including continuous glucose monitoring, may minimize the incidence of hypoglycemia.[19, 20] Future research should match appropriate glycemic targets to various patient populations.[21]

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Conflict of Interest All authors report no conflicts of interest.

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